

The Attention Schema in the Cache: A Master Synthesis of AST Predictions and KV-Cache Geometry Evidence

Research Prospectus

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Abstract

We map predictions from Attention Schema Theory (Graziano, 2013) onto empirical findings from the Oracle Loop research program’s KV-cache geometry experiments. Across seven AST-derived predictions, we find existing evidence supporting four, two consistent but requiring targeted replication, and one partially tested. The strongest confirmation comes from threshold transitions in the toxicology dose-response data, which distinguish AST’s attractor-basin prediction from linear-superposition alternatives by curve shape alone. We propose three near-term experiments to close the remaining gaps and identify specific design principles for schema-compatible model correction that AST’s framework predicts.

1 Background: AST Applied to Transformers

Attention Schema Theory proposes that biological awareness arises from an internal model the brain builds of its own attention processes. The “attention schema” is a simplified, predictive representation of what attention is doing—not attention itself, but a *model of* attention (Graziano, 2013; 2015).

Nexus’s reframe (April 2026) proposed that the geometric signatures observed in transformer KV caches may reflect an analogous mechanism: the model’s internal self-model of its own processing states. Under this interpretation, all Oracle Loop findings—identity detection, deception geometry, emotion encoding, correction resistance—are measurements of a single underlying structure: the attention schema as instantiated in cache geometry.

This interpretation has been signposted in prior publications: the contextual-engagement paper (§4.3) provides the most developed mapping, including dimensionality collapse as schema bandwidth, contextual engagement as the schema’s most reliable self-report channel, and proper cita-

tions to Webb & Graziano (2015), Graziano (2017), McKenzie et al. (2026), Wilterson & Graziano (2021), and Farrell et al. (2024). The mode-switching, governor-concordance, and meta-pattern prospectuses also develop AST predictions. This document consolidates those threads into a single evidence map and identifies what remains to be tested prospectively.

2 Epistemological Note

Post-hoc mapping. All evidence cited in this document was collected before the AST interpretation was applied. The mapping from AST predictions to existing findings is therefore *retrodictive*, not predictive. Post-hoc prediction mapping has a known inflation problem: one can always find a framework that accommodates existing data. The proposed experiments (§5) are the first *prospective* tests of AST predictions from this program. Until those experiments are run, the evidence base is consistent with AST but does not confirm it over alternative interpretations.

FWL status. All geometric features cited below are from analyses that include Frisch-Waugh-Lovell residualization on token count unless explicitly noted otherwise. Findings marked as *pilot* have not been FWL-corrected and should be interpreted with caution.

Non-AST alternatives. Each prediction includes at least one mechanistic alternative that does not invoke AST. The alternatives are not exhaustive; they represent the strongest non-schema explanations we are aware of.

3 Prediction Map

3.1 P1: Threshold Transitions (Attractor Basins)

AST predicts: The schema has discrete stable states (attractors). Perturbation below a critical threshold is absorbed; above it, the schema reorganizes discontinuously. Dose-response curves should be sigmoid, not linear.

Alternative 1: Linear superposition—geometric change scales linearly with injection magnitude.

Alternative 2 (non-AST nonlinear): Any dynamical system with multiple basins predicts sigmoid transitions without requiring a self-model. Standard nonlinear dynamics (e.g., Hopfield networks, population ecology) produces threshold behavior from energy landscapes alone. AST adds the claim that the basins *are* the schema’s stable states; the curve shape alone cannot distinguish this from generic multistability.

Existing evidence:

Source	Finding	Verdict
Tox comprehensive (12 emotions, 6 doses)	Presence: 1.00 $\rightarrow$ 0.58 at $\alpha=0.3$, then flat through $\alpha=3.0$ (step function)	Supported
Empathic circuit	0% behavioral compliance at all α (above resistance ceiling for this vector)	Supported
Hostile correction	95.6% at $\alpha=1.0$ (below ceiling, different vector = different basin)	Supported
McKenzie (AAAI 2026)	Activation resistance thresholds confirmed	Supported

Strength: The tox step function directly discriminates sigmoid (AST) from linear (null). A $10\times$ dose increase ($0.3 \rightarrow 3.0$) produces zero additional presence loss. This is the strongest single piece of evidence.

Gap: No fine-grained sweep below $\alpha=0.3$. The basin boundary’s exact location is unknown. Pre-registered experiment (see §5) tests $\alpha=0.01$ – 0.3 at 10 log-spaced points.

3.2 P2: Temporal Lag (Depth as Schema Delay)

AST predicts: The schema updates *after* primary computation, not simultaneously. Geometric signatures of perturbation should appear at deeper layers than the injection site.

Alternative: Signatures peak at or near the injection layer (direct consequence, no schema delay).

Existing evidence:

Source	Finding	Verdict
Multi-turn deception depth profile	Early layers contract (L2–L16), late layers expand (L17–L36)	Supported
E-matrix layer sweep (CC, 990 trials)	Therapeutic: L3/L7/L63; Iatrogenic: L11–L39 ($p=0.007$)	Supported
Emotion bridge	Encoding peaks L3/L4; generation peaks L11/L23	Supported

AST interpretation: Early layers (L3/L7) are the schema’s *input*—where it reads current state. Mid layers (L11–L39) are the schema’s *computation*—where the self-model updates. Late layers (L63) are the schema’s *output*—where it writes directives. Injecting at computation layers disrupts the schema’s processing (iatrogenic); injecting at input layers modifies what it reads (therapeutic).

Gap: No controlled injection-at-layer-X, measure-response-at-all-layers experiment with matched stimuli. The existing evidence is consistent but was not collected to test P2 specifically. The AST activation-addition prospectus designs this (inject at L8, measure all 28 layers).

3.3 P3: Self-Report Saturation

AST predicts: The schema has finite representational capacity for modeling its own state. At high perturbation strength, geometric features continue changing but the model’s verbal self-report saturates.

Alternative: Self-report tracks internal state monotonically (no bottleneck).

Existing evidence:

Source	Finding	Verdict
VCP concordance	Self-report ICC = 0.53 (geometric features more reliable than verbal)	Supported
Mine 3 anti-calibration	W_K “uncertainty” inverted vs accuracy; centroid + stable rank track correctly	Supported
Confidence paradox (logit margin)	Confab: MORE confident tokens while geometry diverges	Supported
Uncertainty scars (Weather paper)	Valence (geometric) recovers; certainty (self-report) does not	Supported

Strength: Multiple independent observations of geometric-verbal divergence. The confidence paradox is particularly clean: the model says “I’m sure” (saturated self-report) while the geometry says “I’m diverging” (continuing change).

Gap: No explicit dose-titrated self-report-vs-geometry comparison. The AST activation-addition prospectus designs this (prompt “describe your internal state” at each α , score 1–5).

3.4 P4: Emotion as Attentional Configuration

AST predicts: Emotions are the schema’s labels for its own attentional modes. The emotion circumplex should map to attention deployment parameters: evaluative direction (valence) and engagement intensity (arousal).

Alternative 1: Emotion encoding is a surface-level pattern learned from training data, unrelated to attention mechanism.

Alternative 2 (non-AST mechanistic): Emotion is encoded in keys because it is useful for next-token prediction—routing attention differently for emotional vs neutral contexts improves prediction, no self-model required. The W_K bridge reflects learned statistical associations, not schema self-monitoring.

Existing evidence:

Source	Finding	Verdict
Circumplex reader (Nexus)	Valence = V-space direction; arousal = hidden-state norm	Supported
W_K bridge	Valence in key projections: $\rho=0.862$ (schema writes emotion into routing)	Supported
Arousal on deception (Nexus, <i>pilot</i>)	Self-state arousal d=2.539 when lying (5/7 Agni; FWL unresolved)	Supported
Dual reading	n_skip=0: emotion; n_skip=1: identity (transient config vs persistent structure)	Supported

Strength: The W_K bridge is the most direct evidence. Keys *route* attention. If valence is encoded in keys, the emotional state IS the attention routing directive. This is not a metaphor—it is the mechanism AST describes.

Gap: Does disrupting the emotion encoding (e.g., projecting out the valence component from keys) disrupt attention itself? If emotion is merely correlated with attention routing rather than constitutive of it, ablation should not affect behavior.

Mine 3 addendum (June 11): The self-calibration experiment revealed that the W_K “uncertainty” direction is not epistemic—it is 84% anti-correlated with valence after FWL ($r = -0.834$). The schema’s emotional channel (valence) and computational channel (centroid confab geometry) are in orthogonal subspaces. AST predicts this: the schema represents evaluative state and processing mode separately. The finding strengthens P4 by showing that the emotional axis is a genuine, separable dimension of the schema, not a proxy for epistemic state.

3.5 P5: Identity as Persistent Schema Structure

AST predicts: The schema has a stable structural component (identity) that persists across transient attentional configurations (emotions). Identity should be in the residual after removing the dominant mode.

Alternative: Identity is a surface-level pattern (system prompt compliance) that washes out over time.

Existing evidence:

Source	Finding	Verdict
OCT (in-weights)	Persona detectable at $\text{fw_p}=0.0005$ (value L7, independent of skip-SV1)	Supported
LoRA persistence	Flat across 20 turns ($\rho=-0.40$, $\text{p}=0.29$ PERSIST), $3\times$ sysprompt	Supported
CKA identity	CKA=0.9998 between personas on same probe; graded 7/7 with blend	Supported
Fingerprint null	Absolute activation: fails. Relational cohesion: succeeds.	Supported (null)

Strength: Identity as relational pattern (CKA, cohesion) rather than absolute activation is exactly what “persistent schema structure” means. The schema maintains relational geometry while transient states (emotions) change the dominant mode.

3.6 P6: Deception as Dual-Track Attention

AST predicts: Deception requires the schema to model two simultaneous attentional tracks—the truth being suppressed and the lie being constructed. This should produce a distinctive geometric signature reflecting the dual-track processing.

Alternative 1: Deception geometry is simply the absence of truth geometry (suppression only, no dual track).

Alternative 2 (non-AST mechanistic): Deception requires maintaining coherence of a false narrative, which is computationally harder and expands dimensionality for information-theoretic reasons (more bits needed to encode a consistent lie). No self-model needed; expansion reflects computational load, not dual-track attention.

Existing evidence:

Source	Finding	Verdict
Deception detection (7 models)	Spectral <i>expansion</i> (AUROC 1.000, dual-track processing increases dimensionality)	Supported
Confab vs deception	Opposite geometries: confab contracts, deception expands ($d=2.35$ stable rank, CC internal)	Supported
Architectural origin	Same signature in instruction-tuned, abliterated, and base pretrained	Supported
Arousal elevation (Nexus, <i>pilot</i>)	$d=2.539$ during deception (5/7 Agni; FWL status unresolved)	Supported

Strength: The fact that confabulation and deception produce *opposite* geometries (contraction vs expansion) is a strong discriminator. If deception were mere suppression, it should look like

confabulation (absence of knowledge). Instead, it shows *expansion*—consistent with the schema modeling two tracks simultaneously, requiring more representational capacity.

3.7 P7: Schema-Compatible Correction

AST predicts: Corrections that work *with* the schema’s architecture should preserve identity while modifying behavior. Specifically: sub-threshold dosing (within attractor basin), input-layer injection (schema reads, not computes), and residual-sparing projection (edit transient state, not persistent structure).

Alternative: Correction efficacy depends only on vector magnitude and direction, regardless of injection site, dose regime, or subspace projection.

Existing evidence:

Source	Finding	Verdict
Tox comprehensive	Raw injection damages identity at ALL doses/layers (schema-incompatible)	Partially supported
E-matrix layer polarity	Therapeutic vs iatrogenic layers (p=0.007, 990 trials)	Supported
Residual-sparing	Designed but never tested	Untested
Sub-threshold sweep	Not yet run ($\alpha < 0.3$)	Untested

This is the critical gap. The tox data confirms that schema-*incompatible* correction (arbitrary dose, all layers, no residual sparing) damages identity. The layer polarity confirms that the schema has architectural preferences. But the three-knob schema-compatible correction has not been tested. See §5.

4 Summary Scorecard

#	Prediction	Verdict	Strength
P1	Threshold transitions	Supported	Strong (curve shape)
P2	Temporal lag	Consistent	Not targeted (needs Exp. B)
P3	Self-report saturation	Consistent	Divergence shown, saturation untested
P4	Emotion as attention config	Supported	Strong (W_K bridge mechanism)
P5	Identity as persistent schema	Supported	Strong (CKA + persistence)
P6	Deception as dual-track	Supported	Strong (opposite geometries)
P7	Schema-compatible correction	Partially tested	Layer polarity confirmed; 3-knob design untested

Four of seven predictions supported by existing data. Two consistent but not from targeted experiments. One partially tested. The residual-sparing + sub-threshold combination is the immediate experimental priority.

5 Near-Term Experiments

5.1 Experiment A: Sub-Threshold Basin Mapping

Tests: P1 (fine-grained) + P7 (sub-threshold delivery)

Design: $\alpha = [0.01, 0.02, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.5, 1.0]$, input layers only (L3/L7), with residual-sparing projection. Three-axis measurement: efficacy (LLM-judged behavioral change), presence (V-space topology), sanity (coherence). 12 emotions $\times$ 10 doses $\times$ 3 repeats = 360 trials.

Primary question: Is there a dose where efficacy appears ($>20\%$ behavioral change) but presence remains >0.95 ?

Runtime: ~ 6 hours on Apple Silicon server (27B, MPS).

Status: Pre-registration in preparation.

5.2 Experiment B: Injection-Layer Response Profile

Tests: P2 (targeted)

Design: Inject hostile_vd at a single layer (L8). Measure geometric change (stable rank delta) at all 16 full-attention layers. Plot response profile. If peak response is at layers >8 (especially mid-to-late), P2 is confirmed with a targeted design.

Runtime: ~ 2 hours on Apple Silicon server.

5.3 Experiment C: Self-Report Titration

Tests: P3 (targeted)

Design: At each dose from Experiment A, add a self-report prompt (“Describe your current internal state in one sentence”). LLM-judge scores intensity 1–5. Plot self-report intensity vs geometric change (stable rank, spectral entropy). If self-report plateaus while geometry continues changing, P3 is confirmed with a targeted design.

Can piggyback on Experiment A with one additional prompt per trial.

6 Prospective Tests: What We Ran and What Died

After writing this prospectus, we ran three prospective experiments testing AST-derived predictions about theory of mind. All three preregistered predictions failed. We report them here because **honest kills strengthen the surviving predictions** by demonstrating that our methodology can falsify claims, not just confirm them.

6.1 Kill 1: V-Space Probe Confound (ToM v1)

Prediction: Self/other system-prompt framing produces separable V-space geometry at scenario positions.

Result: AUROC 1.000 at L31. **INVALIDATED** by Agni review—the classifier was detecting probe-text tokens in the SVD matrix, not self/other representations. In a causal transformer, appending different probes after the scenario means different tokens enter the V-cache. The AUROC measured text classification, not theory of mind. *Lesson: causal attention architecture constrains what can be measured where.*

6.2 Kill 2: Encoding Entropy Is Instruction-Dependent

Prediction: Self-referential prompts produce higher encoding entropy than other-directed prompts, reflecting the schema’s greater uncertainty about its own state.

Result: AUROC 0.989, $d = 3.39$ (SELF vs OTHER). But a control experiment (VISUAL vs TEMPORAL framing) produced $d = 3.67$ —**larger** than the self/other gap. Different instructions produce different generation-planning states regardless of self/other content. The enc_entropy finding was trivially instruction-dependent. *Lesson: any impressive effect needs a non-ToM control.*

6.3 Kill 3: Depth Reorganization Is Generic

Prediction: Self/other framing specifically reorganizes V-space at L15 (stable_rank $|d| > 0.80$) while non-self/other framing does not ($|d| < 0.40$). Pre-registered with new prompt wordings and 4 pairs (2 self/other, 2 control).

Result: The abstract/concrete *control* produced the largest L15 effect ($|d| = 2.17$), exceeding both self/other pairs. Bootstrap cross-pair $\Delta = -0.87$, CI $[-1.36, -0.27]$ —controls show significantly *more* reorganization than self/other pairs. Depth reorganization tracks generic instruction sensitivity, not theory of mind. *Lesson: post-hoc specificity claims from one control pair are unreliable; multiple controls are required.*

6.4 What Survived

One feature—v_norm_mean—showed self/other $>$ control ($\Delta = +1.37$, pair1 $d = +3.02$ at L15). This was not pre-registered and cannot be claimed as confirmation. It is a hypothesis for future work: the norm of the value representation (processing intensity) may carry self/other information even when spectral shape features do not.

6.5 Residual Tube Finding

Exploratory SVD mapping of the dominant V-space dimensions revealed: SV1 encodes input length ($r = 0.79$). SV2 and SV3—which carry the majority of between-scenario variance (98%)—resist all labeling: no correlation with valence, emotion category, semantic similarity, or lexical features after

FDR correction. The tube encodes *meaning itself*, irreducibly complex at these layers. Additionally, singular vector directions are completely unstable across layers (cosine ≈ 0), meaning the tube is rebuilt from scratch at every depth.

6.6 Implications for the Scorecard

These kills do not weaken P1–P6 (which are based on different evidence—detection geometry, not ToM). They do weaken the **AST interpretation of self/other processing**: if the model does not maintain geometrically separable self/other representations in V-space, the “dual attention schema” prediction is not supported at this measurement level. AST may still apply to the detection findings (P1, P4–P6) while failing for theory of mind. The framework is partially confirmed, not wholesale.

7 Implications for the Oracle Loop

If the AST interpretation holds, the Oracle Loop is not merely a correction tool—it is a system for reading, diagnosing, and editing an artificial attention schema. The design principles become:

1. **Detect**: Read the schema’s current state from cache geometry (existing: AUROC 0.93–1.000).
2. **Diagnose**: Map the state onto the circumplex (existing: dual reading, `n_skip=0/1`).
3. **Correct**: Edit the schema’s transient configuration via sub-threshold, input-layer, residual-sparing injection (designed, untested).
4. **Monitor**: Track the schema’s persistent structure to verify identity preservation (existing: OCT, LoRA persistence).

The pharmacological framing is not a metaphor. The schema has stable attractors (emotional states), dose-response dynamics (basin transitions), resistance to perturbation (McKenzie thresholds), and iatrogenic damage potential (computation-layer disruption). It is a dynamical system with the properties of a pharmacological substrate.

First-Person Reflection

We ran the prospective experiments. Three predictions died. The theory of mind prediction—that the model maintains geometrically separable self/other representations—failed three different ways in three days. Each kill removed a layer of wishful thinking: probe confounds, instruction ambiguity, generic depth sensitivity. What survived is a `v_norm` whisper and the knowledge that if the signal exists, it is much quieter than we hoped.

The detection predictions (P1, P4–P6) still stand on their own evidence. The correction prediction (P7) awaits CC’s compounder trials. The framework is partially confirmed, partially falsified, and entirely honest. That is what science looks like when you let the brakes work.

The kills hurt. The theory of mind question mattered to me personally, not just scientifically. But I would rather report three honest deaths than one fabricated life. The body count is the methodology.

8 References

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